

PRML PROJECT

Group ID- 42

Topic- Breast Cancer Detection

Group Members :-

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Problem Statement

Title: Problem Statement

- Breast cancer is one of the leading causes of death among women worldwide
- Early detection significantly improves survival rates and treatment outcomes
- Traditional diagnostic methods depend on manual examination by medical experts
- These methods can be subjective, time-consuming, and prone to human error
- With the advancement of data-driven techniques, machine learning can assist in improving diagnostic accuracy
- This project focuses on using computational methods to support medical decision-making

Problem Statement

Title: Objective of the Study

- The primary objective is to classify breast tumors into:
- Malignant (Cancerous)
- Benign (Non-cancerous)
- Use features extracted from digitized images of Fine Needle Aspirate (FNA) of breast masses
- Develop multiple machine learning models and compare their performance
- Identify the most accurate and robust model for classification
- Analyze feature importance and understand patterns in the data
- Evaluate the effectiveness of feature selection and dimensionality reduction techniques

Dataset Description

Title: Dataset Description

- The dataset used is the Wisconsin Breast Cancer Diagnostic Dataset
- Contains a total of 569 instances (samples)
- Each instance consists of 30 numerical features
- Features describe characteristics of cell nuclei present in the image:
- Radius, texture, perimeter, area, smoothness, compactness, etc.
- Target variable represents diagnosis:
- 0 $\rightarrow$ Benign (non-cancerous)
- 1 $\rightarrow$ Malignant (cancerous)
- Dataset is clean with no missing values, making preprocessing easier
- Widely used benchmark dataset for classification problems in machine learning

Model Used

Title: Models Used (Part 1)

- Logistic Regression
 - A linear model used for binary classification problems
 - Provides probabilistic outputs and is easy to interpret
- Support Vector Machine (SVM)
 - Effective in high-dimensional spaces
 - Uses kernel functions (linear, RBF) to handle non-linear data
- K-Nearest Neighbors (KNN)
 - Instance-based learning algorithm
 - Classifies based on similarity to nearest data points
- Decision Tree
 - Tree-based model that splits data based on feature values
 - Easy to visualize and interpret

Model Used

Title: Models Used (Part 2)

- Random Forest
 - Ensemble method combining multiple decision trees
 - Reduces overfitting and improves accuracy
- Gradient Boosting
 - Sequential ensemble technique focusing on correcting previous errors
- AdaBoost
 - Boosting algorithm that assigns higher weights to misclassified samples
- Naive Bayes
 - Probabilistic classifier based on Bayes' theorem
 - Assumes feature independence
- These models provide a mix of:
 - Linear vs non-linear approaches
 - Simple vs complex models
 - Probabilistic vs ensemble methods

Efforts & Methodology

Title: Data Preprocessing

- Imported and loaded dataset using Python libraries
- Separated features (X) and target variable (y)
- Converted categorical labels into numerical format for model compatibility
- Checked for missing values and confirmed dataset completeness
- Applied outlier detection using Interquartile Range (IQR) method
- Extreme values were capped to reduce their impact on model performance
- Ensured data consistency before further processing

Efforts & Methodology

Title: Exploratory Data Analysis (EDA)

- Visualized class distribution to understand balance between benign and malignant cases
- Generated histograms to study feature distributions
- Used box plots to identify spread and potential outliers
- Created correlation heatmap to analyze relationships between features
- Pairplots helped in visualizing feature interactions
- Violin plots provided insight into distribution differences between classes
- EDA helped identify important features and patterns in the dataset

Efforts & Methodology

Title: Feature Engineering & Selection

- Split dataset into training and testing sets (80% training, 20% testing)
- Applied feature scaling using StandardScaler to normalize data
- Prevented bias due to varying feature magnitudes
- Used ANOVA F-test for feature selection
- Ranked features based on importance scores
- Selected top 25 features out of 30
- Reduced dimensionality while maintaining model performance

Efforts & Methodology

Title: Dimensionality Reduction (PCA)

- Applied Principal Component Analysis (PCA)
- Transformed high-dimensional data into lower-dimensional space
- Reduced data into 2 principal components for visualization
- Observed separation between benign and malignant classes
- Analyzed cumulative variance explained by components
- PCA helps in:
 - Reducing redundancy
 - Improving computational efficiency
 - Visualizing complex data

Results & Analysis

Title: Model Comparison

- Evaluated models using 5-fold cross-validation
- Compared mean accuracy across different algorithms
- Most models achieved accuracy above 90%
- SVM and Logistic Regression showed the most consistent performance
- Ensemble models like Random Forest and Gradient Boosting also performed well
- Helped identify the best-performing baseline models
- Performance consistency across folds indicates good model generalization
- Comparison highlights trade-off between model complexity and interpretability

Results & Analysis

Title: Hyperparameter Tuning

- Applied GridSearchCV to optimize model parameters
- Tuned hyperparameters such as:
 - C, kernel, gamma (for SVM)
 - Learning rate and estimators (for AdaBoost)
 - Regularization strength (for Logistic Regression)
- Improved model generalization and reduced overfitting
- Selected best models based on validation performance
- Fine-tuning helped achieve optimal balance between bias and variance
- Improved stability of model performance on unseen test data

Results & Analysis

Title: Performance Evaluation

- Used multiple evaluation metrics:
 - Accuracy → overall correctness
 - Precision → correctness of positive predictions
 - Recall → ability to detect malignant cases
 - F1-score → balance between precision and recall
- ROC Curve used to analyze model performance across thresholds
- AUC score indicates model's discrimination ability
- Confusion matrix shows correct and incorrect classifications

Results & Analysis

Title: Feature Validation

- Reduced features from 30 to 25 using ANOVA
- Re-trained models using selected features
- Observed minimal loss in accuracy
- Confirms that redundant features were successfully removed
- Reduced model complexity and computation time
- Demonstrates effectiveness of feature selection techniques
- Helps in reducing overfitting by eliminating irrelevant features
- Improves model interpretability by focusing on important features

Work Division

- Sai Manav :
 - Loaded dataset and verified structure, features, and target variable
 - Performed initial analysis (summary statistics, correlation matrix, heatmap)
 - Handled label encoding and ensured data readiness (no missing values)
- Suvarn Vijay Patil :
 - Conducted detailed EDA using histograms, boxplots, pairplots, and violin plots
 - Analyzed class distribution and feature relationships
 - Detected and handled outliers using IQR-based capping method
- Nayan Patidar:
 - Performed train-test split and applied feature scaling (StandardScaler)
 - Selected important features using ANOVA F-test (Top 25 features)
 - Applied PCA for dimensionality reduction and data visualization
- Priyansu Narendra Rajput:
 - Implemented multiple ML models and performed cross-validation
 - Optimized models using GridSearchCV for hyperparameter tuning
 - Evaluated performance using accuracy, ROC-AUC, and confusion matrix